1) Functional Requirements

Student Functions
=> User Registration /Login
-> Join Assessment
-> view Questions
-> write & submit code
-> code Execution
-> view submission status
-> view Results

Admin Functions
-> create Assessments
-> Manage Questions
-> manage Test cases
-> Review Results

2) Non- Functional Requirements
-> Availability
-> Scalability to support thousands of students
-> API Response Time
-> submission Handling.
-> security
-> Reliability
-> Data Durability
-> consistency
-> maintainability
-> Fault Tolerant

Assumptions

User :-

→ The platform has approximately 100,00 registered user

→ 10,000 students write exam same time

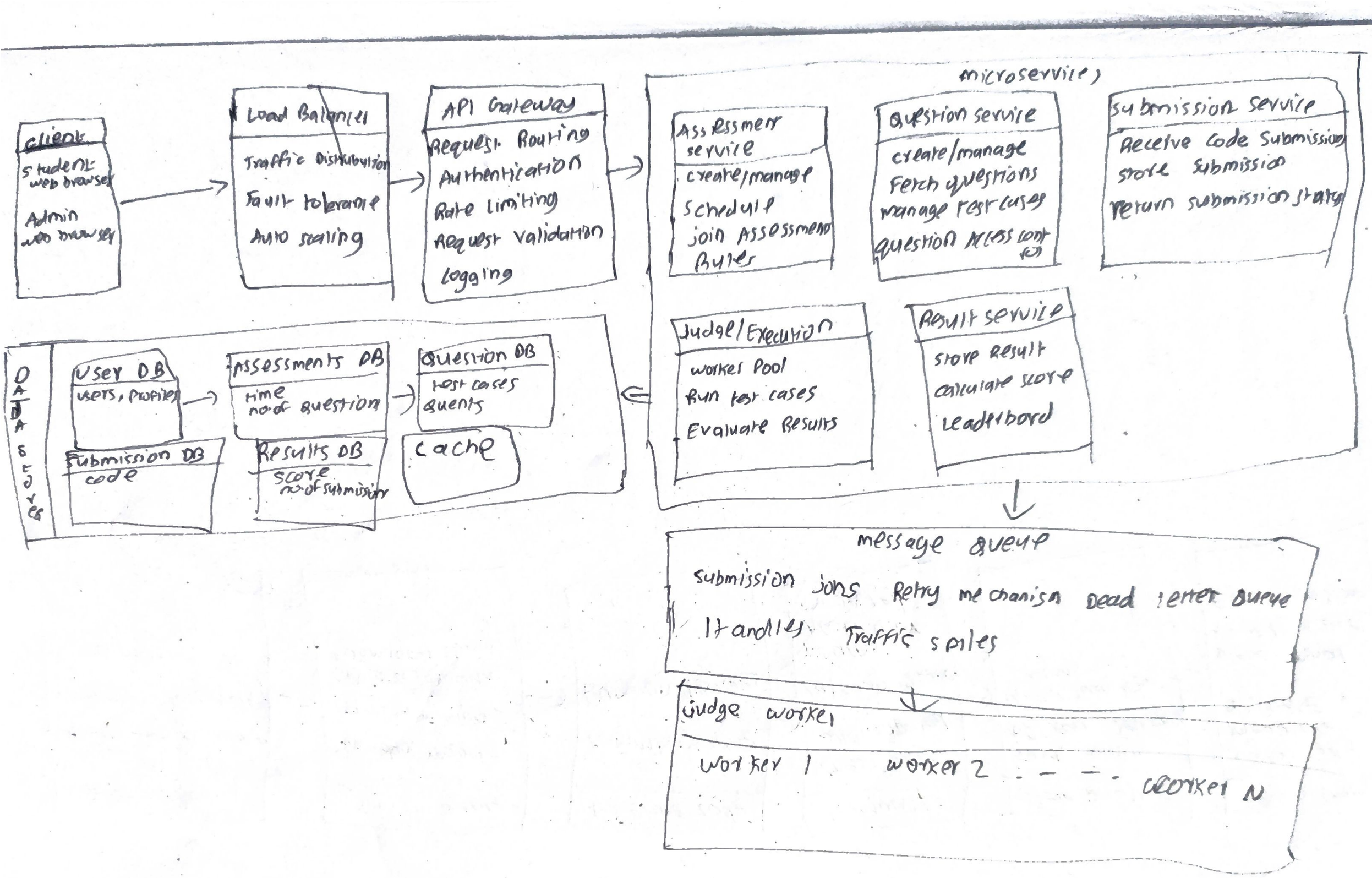
Assessment:

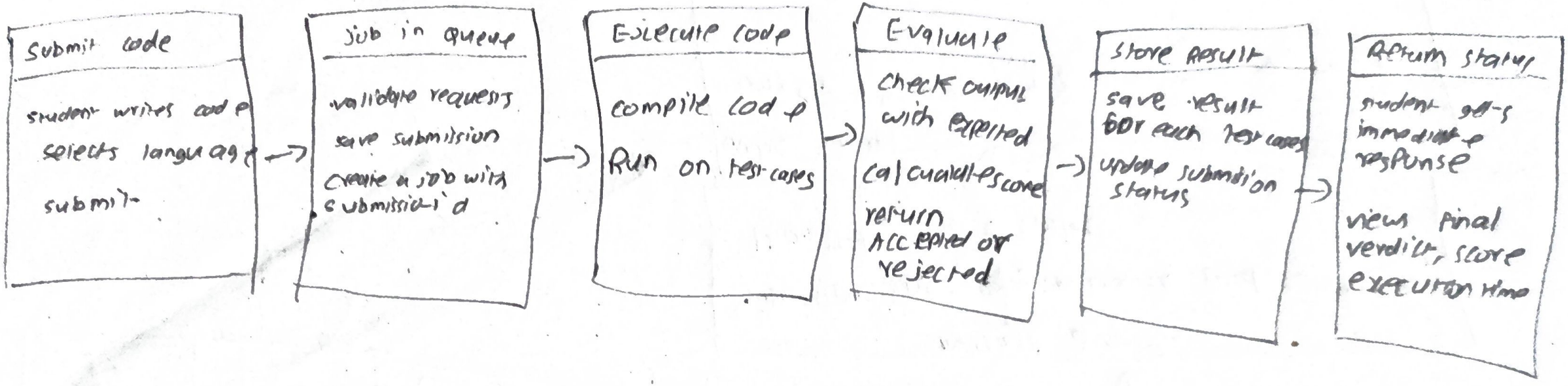
→ 30-50 questions

→ time of exam can be 30 - 180 min

→ Public test cases & hidden test cases

microservice


Client
Student
web browser
Admin
web browser
Load Balancer
Traffic distribution
Fault tolerance
Auto scaling
API Gateway
Request Routing
Authentication
Rate Limiting
Request Validation
Logging
microservice)
Assessment servvice
create/manage
schedule
join Assessment
Rules
Question servvice
create/manage
Fetch questions
manage test cases
question Access cont
Submission servvice
Receive code Submission
store Submission
Return submission status
Judge/Execution
worker Pool
Run test cases
Evaluate Results
Result servvice
store result
calculate score
Leaderbord
DATABASE
User DB
users, Profiles
Assessments DB
time
no of question
Question DB
test cases
quents
Submission DB
code
Results DB
score
no of submissions
cache
message queue
submission jons
Retry mechanism
Dead letter queue
Handley
Traffic spikes
Judge worker
worker 1
worker 2
worker N


Submit code
student writes code
selects language
submit
Job in queue
validate request
save submission
create a job with submission_id
Execute code
compile code
Run on test cases
Evaluate
check output with expected
calculate score
return Accepted or rejected
Store result
save result for each test case
update submission status
Return status
student gets immediate response
view final verdict, score execution time